

Shift AI Technology Ltd — Diligence Fact Pack + Technical Whitepaper

Updated: 2026-09-09
Sources: Companies House; Jack Rockell answers; BioSense repo; FibreCRM/Braid monorepo; Deep Space Learning docs; NVIDIA DGX SETUP folder.
Tone: facts only.

Part 1 — Diligence answers

1. Company details

Field	Value
Legal name	SHIFT AI TECHNOLOGY LTD
Company number	17159725
Status	Active
Type	Private limited company
Incorporated	15 April 2026
Registered office	Unit 14, The Maltings Industrial Estate, Hall Road, Southminster, Essex, CM0 7EQ, United Kingdom
SIC	62090; 63110
Directors	Darren Bishop; Jack Rockell
Trading as	Shift AI Tech
Domain	https://shiftaitech.com
Contact	jack@shiftaitech.com · partnerships@shiftaitech.com

2. People

Role	Count	Notes
Co-founders	2	One commercial / one technical (dev)
Subcontractors	2	Active
Total working in/for Shift now	4	
Developers / technical	~3	Technical co-founder + 2 subcontractors (confirm if commercial co-founder also codes)

3. Funding

- **No outside investment.** Fully self-funded / bootstrapped for Shift Ltd.
- Historical story capital (website): ~\$650k spent on Bitcoin prediction models (2024) linked to Infinite Point Capital narrative; **separate from** CH incorporation funding status.

4. AWS credits

- **No AWS credits** (not that Jack knows of).
- AWS **is** used in production (see infra below), paid as normal cloud spend.

5. Accelerators / incubators / VC–angel affiliations

- **None.**

6. NVIDIA / GPU

- **Own NVIDIA DGX Spark** already (GB10 Grace Blackwell class, documented locally as `spark-*`).
- Intended use: **training AI models**; looking to expand NVIDIA usage.
- Documented DGX workloads today (local ops folder): PyTorch / transformers; FLUX.1 image + LoRA; ChromaDB + sentence-transformers RAG for trading/quant tooling; bot/trading stacks. Deep Space video path is **not** DGX-dependent (HeyGen SaaS + Remotion CPU).

7. Products Shift owns (Jack: own all)

Product	Live surface	Status
BioSense	bio-sense.ai / Cloud Run	Owned health intelligence SaaS
Braid (AML / CRA skins)	usebraid.ai · cra.fibrecrm.ai · aml.shiftaitech.com	Owned AI AML risk-assessment platform (built for FibreCRM commercial relationship; Jack confirms Shift owns)
Deep Space Learning (EduStream)	Deep Space Commodities course platform	Owned AI-avatar L&D product
Other portfolio items on marketing site (screener, churn, nutrition)	Case/portfolio	Jack: own all — treat as Shift IP unless later corrected

Website client builds (Red Cow, Mixa, etc.) are delivery work, not SaaS products.

8. Traction

- **Revenue to date:** ~£10k made already (confirm currency if not GBP).
- **Ongoing live / active:** BioSense, Braid AML, Deep Space.
- Partnership: **RAMDVG** (tokenisation) for joint AI go-to-market (esp. legal / asset).
- No formal pitch deck currently.

9. Decks

- **No pitch decks.**
- Capability materials: websites flyer; this whitepaper; product READMEs / handover docs in repos.

10. AWS usage (from code, not a billing export)

Area	Services observed
Shift marketing site forms	API Gateway eu-west-2
Braid / CRA (FibreCRM AWS acct 361226991768, eu-west-2)	ECS Fargate, ECR, S3, CloudFront, RDS PostgreSQL Multi-AZ, (CloudFormation under infrastructure/)

BioSense	Primary = GCP Cloud Run + Cloud SQL (not AWS). R2 (Cloudflare) for PDFs. Optional Vercel cron defs exist but prod narrative is GCP.
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Monthly spend: not in repos. **ASK for AWS/GCP bill totals if application needs a number.**

Part 2 — Technical whitepaper (detail dump)

A. BioSense — personal health intelligence

Repo: Shift-ai-tech-ORG/BioSense-App · local /Users/jackrockell/Desktop/biosense

Live (README): Cloud Run europe-west2 , project naming shift-biosense

Stack

- Next.js **16.2.4** App Router, React 19, TypeScript, Tailwind v4
- Prisma **7** + PostgreSQL 15 (@prisma/adapters)
- NextAuth v4 (Credentials + JWT, bcrypt cost 12)
- Anthropic SDK · model **claude-sonnet-4-5**
- Stripe subscriptions (AED 149/mo, AED 1499/yr in code)
- Cloudflare R2 for blood PDF storage
- Resend (email helpers present)
- Docker Node 22 · Cloud Build → Artifact Registry → Cloud Run (1Gi, max 10)

Architecture

```
Browser (App Router)
  → NextAuth session
  → Route Handlers /api/*
  → Prisma → Cloud SQL Postgres
  → Claude (blood analysis, chat, weekly/monthly reports)
  → R2 (PDFs)
  → Stripe (billing webhooks)
  → Cloud Scheduler → /api/cron/* (x-cron-secret)
```

LLM / agents

- **Not multi-agent.** Single-call Claude wrapper with retries on 529/429.
- Surfaces: blood PDF analysis (JSON biomarkers T1/T2/T3); “Ask Anything” chat with DB context stuffing; weekly/monthly report JSON.
- App 5 medical-language policy in system prompts (no diagnosis claims).
- Deterministic (non-LLM): health score, bio-age heuristic, Pearson lag patterns, anomaly thresholds.

RAG

- **None.** Personalisation = SQL context into prompt, not vector retrieval.

Data model (highlights)

User, DailyCheckin, HealthScore, BloodResult, BiologicalAge, WearableSync, ChatMessage, WeeklyReport, MonthlyReport, Pattern, Consent, NotificationLog, Stripe fields on User.

Integrations

Oura / Whoop / Garmin OAuth; Apple Health JSON upload; Samsung auth stub; Stripe Checkout + Portal + webhook; R2; Resend.

Security

JWT sessions 30d; per-route session checks (no Next middleware file); cron secret header; Stripe signature verify; consent versions + IP; account export + hard delete (PDPL-oriented); Cloud Run `--allow-unauthenticated` with app-level auth.

Gaps / incomplete (honest)

Web push send path unwired; lag-analysis writer / bio-age writer not fully wired from routes; wearable metric sync after OAuth limited; some README “done” items partial.

B. Braid / CRA / AML — AI risk assessments

Canonical repo: `FibreCRM/AI-Risk-Assessments` · local `/Users/jackrockell/Desktop/ACCOUNTING SOFTWARE`
Skins (same codebase):

- Braid → `usebraid.ai` (self-serve, `product_type=standalone`)
- CRA → `cra.fibrecrm.ai` (FibreCRM enterprise)
- AML → `aml.shiftaitech.com` (legacy Hetzner)

Built by Shift AI for FibreCRM commercial channel; Jack confirms Shift owns the product IP.

Stack

- Backend: Python 3.13, Flask 3.1, Gunicorn, Flask-JWT-Extended, SQLAlchemy, WeasyPrint, Sentry
- Frontend: React 19, Vite 8, TypeScript, Tailwind v4, React Router 7
- AI: **Google Gemini 2.5 Flash + Pro on Vertex AI** (`europe-west1` when Vertex enabled)
- Search: **Tavily** (adverse media + enrichment)
- QA: deterministic `trial/qa_checker.py` (“Python decides score, AI decides words”)
- Payments: Stripe (GBP) checkout, subscriptions, top-ups, portal; webhook idempotency via `StripeEventLog`

Infrastructure

Env	Host	DB	Compute
AML	Hetzner VPS	SQLite	Nginx + Gunicorn
CRA	AWS eu-west-2	RDS Postgres Multi-AZ	ECS Fargate
Braid	Same AWS account	RDS Postgres	ECS Fargate + S3/CloudFront

Pipeline

Firm submits company/individual → Companies House / Charity / web → FCDO sanctions + OpenSanctions PEP → Tavily adverse media → Gemini report → QA rules → HTML/PDF → UI and/or Make.com webhook (enterprise).

Auth / tenancy

JWT (8h); TOTP MFA; roles user/partner/admin + firm admin; tenant = `Firm` ; API keys SHA-256 for enterprise tasks; Microsoft Entra SSO per firm; affiliate realm JWT.

Security posture (documented)

Internal audits May/July 2026; customer protection HTML; **not CREST**; July follow-up recommended independent pen test. FibreCRM org holds Cyber Essentials / IASME (FibreCRM corporate, not automatically Shift Ltd accreditation).

C. Deep Space Learning (EduStream)

Repo: `Shift-ai-tech-ORG/deep-space-learning` (private)
Product: Netflix-style course platform for *Deep Space Commodities and the New Space Economy*.

Stack

Next.js 15, React 19, TypeScript, Tailwind · Supabase (Postgres + Auth) · Stripe one-time GBP unlock · OpenAI **GPT-4o-mini** tutor · Cloudflare R2 video (`deep-space-course`) · Remotion 4 · HeyGen avatar API · Vercel deploy target.

Avatar / video pipeline

```
Lesson markdown
→ HeyGen (talking-head avatar + voice + captions)
→ Unsplash B-roll
→ Remotion composite (avatar + B-roll + lower-thirds)
→ ffmpeg
→ R2 upload
→ SQL points lessons.video_url → r2:...
→ GPT tutor prompts from captions
```

Scale: 12 chapters / 73 lessons; Chapter 1 documented live end-to-end (as of mid-2026 handoff notes).

GPU

Not required for Deep Space video (HeyGen SaaS + Remotion CPU). Separate from DGX training box.

D. NVIDIA DGX Spark (on-prem)

Documented under `/Users/jackrockell/Desktop/NVIDIA DGX SETUP` :

- NVIDIA DGX Spark / **GB10**
- PyTorch, transformers, accelerate, bitsandbytes
- FLUX.1-dev / schnell + DreamBooth LoRA training
- ChromaDB + sentence-transformers RAG for quant/trading KB
- Trading bot / optimizer workloads via screen/rsync

Roadmap intent (Jack): expand NVIDIA usage for **model training**.

E. Cross-cutting IP / deployment / roadmap notes

Proprietary IP (as claimed)

- BioSense application + health scoring / Claude App-5 prompt regime
- Braid/CRA/AML assessment engine (`trial/`), QA rules, multi-skin SaaS
- Deep Space course platform + avatar pipeline tooling
- DGX local training / FLUX / RAG tooling

Cloud footprint summary

- **GCP:** BioSense (Cloud Run, Cloud SQL, Secret Manager, Scheduler, Artifact Registry)
- **AWS:** Braid/CRA (ECS, RDS, S3, CloudFront, ECR); Shift site forms API Gateway
- **Cloudflare:** R2 (BioSense PDFs; Deep Space video)
- **Supabase / Vercel / Hetzner:** Deep Space / legacy AML

Near-term roadmap (inferred + Jack intent)

1. Keep BioSense, Braid, Deep Space live and shipping
 2. Scale NVIDIA / DGX for training
 3. RAMDVG joint AI for legal / tokenised assets
 4. Formalise pen test cadence for Braid (ICAEW / FibreCRM thread)
 5. Produce pitch deck when needed (none today)
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F. Still missing if applications demand it

1. Exact **monthly AWS + GCP spend** (billing console)
 2. Whether revenue ~10k is **GBP** and period (LTD lifetime vs last 12 months)
 3. Named customer logos / contracts we can disclose
 4. Pen-test vendor booking for Braid (Cool Waters / clone CRA discussed)
 5. Confirm commercial co-founder coding split if they want "X of Y are developers" exact
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One-paragraph company summary (factual)

Shift AI Technology Ltd (17159725, incorporated 15 April 2026, Southminster) is a bootstrapped UK AI product studio run by two co-founders plus two subcontractors (~3 technical). It owns and operates BioSense (Next.js/Claude/GCP), the Braid/CRA/AML assessment platform (Flask/React/Vertex Gemini on AWS and Hetzner), and Deep Space Learning (Next.js/HeyGen/Supabase/R2). Traction: ~£10k revenue to date with those products ongoing; RAMDVG partnership for tokenisation + AI. No outside equity, no accelerators, no AWS credits known. On-prem NVIDIA DGX Spark is used / planned for model training; looking to deepen NVIDIA usage. No pitch deck yet; technical detail lives in product repos and this document.